AS4545

UNDERGRADUATE RESEARCH PROJECT

Self-Sensing Glass-Fibre / CNT Composites under Flexural Loading: Failure Forecasting from Electrical Resistance

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1 Introduction

This report studies a glass-fibre composite in which carbon nanotubes (CNTs) are dispersed in the polymer matrix, and asks whether the material's own electrical resistance can be used to detect damage and warn of failure during bending. The first half of the report explains the material, the test, and the structure of the data. The second half describes the models that were built, the way they were evaluated on bars they had never seen, and the performance of the final early-warning alarm.

In summary, a model that reads the resistance signal predicts the remaining deflection a bar can sustain before failure to within about 1.5 mm, and a calibrated alarm built on that prediction fires before failure on every held-out bar, with a median warning of about 74 seconds and no false alarms.

The objectives of the project were:

- to characterise how the four bars (S1 to S4) behaved, mechanically and electrically, as they were bent to failure;
- to explain the physical origin of the resistance response;
- to define a prediction target that is well posed despite the small number of bars and the differences between them;
- to build and honestly evaluate models that forecast failure from the signal; and
- to convert the best model into a calibrated and usable early-warning alarm.

2 Background

2.1 Why a self-sensing material is useful

Composites carry load in many critical structures, such as aircraft panels and wind-turbine blades. A practical difficulty is that damage usually begins inside the material, as micro-cracks, where it cannot be seen. Detecting it normally requires the structure to be taken out of service and inspected, which is slow and costly. A self-sensing material addresses this directly: the material itself produces an electrical signal that changes as it is strained or damaged, so its condition can in principle be monitored while it remains in use.

2.2 How the CNT network senses strain and damage

Glass fibre does not conduct electricity. The conductivity of the composite comes from the carbon nanotubes dispersed in the polymer matrix. Above a critical concentration, the percolation threshold, the nanotubes form a connected network through which current can flow. A significant part of this conduction occurs by electron tunnelling across small gaps between neighbouring nanotubes.

The useful property is that the resistance of this network changes when the material is strained. It is tracked here as the fractional change in resistance,

$$\frac{\Delta R}{R_0} = \frac{R - R_0}{R_0}, \tag{1}$$

where R_0 is the resistance before any load is applied. The sensitivity of the material to a strain ε is described by the gauge factor,

$$\mathrm{GF} = \frac{\Delta R/R_0}{\varepsilon}. \tag{2}$$

The high sensitivity is a consequence of tunnelling. The resistance across a gap grows approximately exponentially with the gap width d,

$$R_{\text{tunnel}} \propto \exp(\gamma\, d), \qquad \gamma = \frac{4\pi}{h}\sqrt{2m\lambda}, \tag{3}$$

where h is Planck's constant, m the electron mass and λ the barrier height. Because this dependence is exponential, even a small strain that widens the gaps produces a large change in resistance. As loading continues, micro-cracks in the matrix sever conducting paths and raise the resistance further, sometimes in sudden steps. Since the glass fibres carry no current, the resistance reports essentially on the matrix, which is where damage initiates.

2.3 The three-point bending test

The specimens were tested in three-point bending. The bar rests on two supports a distance L apart, and a central nose presses down with a load P, stretching the lower face and compressing the upper one. For a rectangular bar of width b and thickness h, the maximum surface stress and strain at mid-span are

$$\sigma_f = \frac{3PL}{2bh^2}, \qquad \varepsilon_f = \frac{6Dh}{L^2}, \tag{4}$$

where D is the mid-span deflection. The flexural modulus follows from the initial slope m of the load against deflection curve,

$$E_f = \frac{L^3 m}{4bh^3}. \tag{5}$$

The test is displacement controlled: the nose advances at a constant speed and the load is recorded until the bar fails, at which point the load reaches a peak and then drops. These are the standard flexural relations of ASTM D790.

3 Experimental data

3.1 Test setup

The tests were performed in the laboratory and the recorded data was provided for analysis. Two instruments ran simultaneously during each test, as sketched in Figure 1. A universal testing machine (UTM) bent the bar and recorded the load and the cross-head deflection. At the same time, a Keithley source-meter applied a steady voltage of about 20 V across the bar and measured its resistance through a two-wire connection. Both instruments sampled at about 12 Hz.

3.2 Recorded quantities

The procedure was the same for each of the four bars. The bar was placed on the two supports and its initial resistance R_0 was noted. The nose then pressed down at a constant speed, bending the bar until failure. Throughout the test the UTM saved time, load in kN and deflection in mm, while the source-meter saved resistance in ohms against its own clock, both at about 12 Hz.

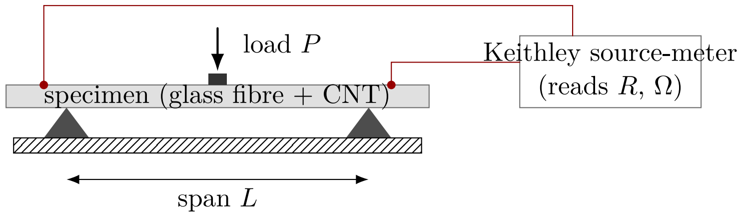

Figure 1: The three-point bending test with simultaneous resistance measurement. The UTM bends the bar to failure while the source-meter reads the resistance of the internal CNT network.

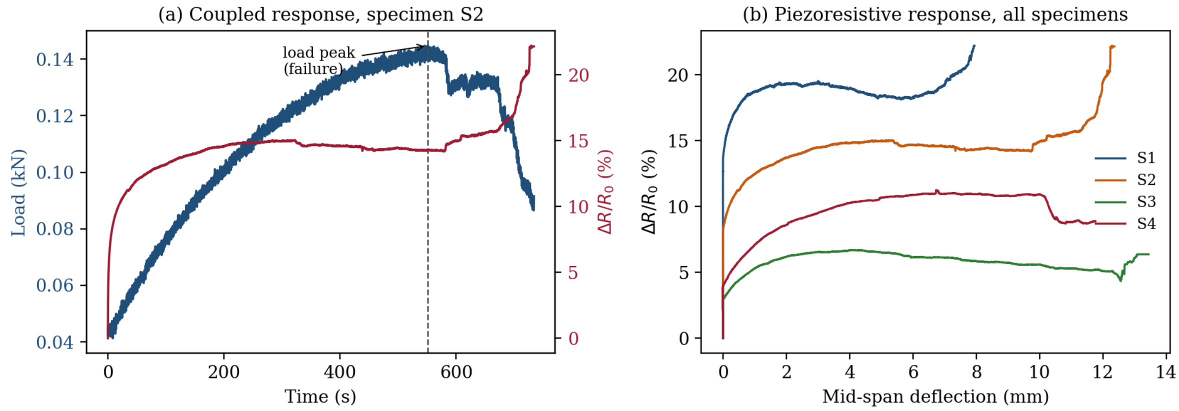

Figure 2: (a) Load and fractional resistance change over time for bar S2: the load rises, peaks at failure, then drops, while the resistance rises and then levels off. (b) Fractional resistance change against deflection for all four bars.

In post-processing the two records were placed on a common time base and the resistance was converted to the fractional change $\Delta R/R_0$, so that bars with different starting resistances could be compared on equal terms.

4 Observations

Figure 2(a) shows the aligned load and resistance for one bar, and Figure 2(b) shows the resistance change against deflection for all four. Table 1 lists the main quantities, computed by the data pipeline described in Section 5. The principal observations are as follows.

- **The signals move together.** As the bar bends, the load rises and peaks at failure before dropping, the deflection grows steadily, and the resistance climbs. The mechanical and electrical responses are clearly linked.
- **The resistance responds early.** It completes most of its rise at the start of the test, reaching about 80% of its pre-peak value within the first 10 to 20% of the test. This is consistent with the tunnelling mechanism, which is most sensitive to the first strains and the earliest micro-damage.
- **A plateau is followed by features near failure.** Through the middle of the test the resistance is nearly flat, and near the load peak it shows distinct steps. It often continues to rise after the load peak, as the crack opens further while the bar is pushed on.
- **The baseline resistance varies widely.** The starting values range from about 7 to

Table 1: Summary quantities for the four bars, from the reproducible pipeline. D_{peak} is the deflection at the load peak (the failure point).

Bar	R_0 (MΩ)	Peak load (kN)	D_{peak} (mm)	Max. $\Delta R/R_0$ (%)	Loading rate (mm/s)	Duration (s)
S1	52.4	0.180	7.36	14.8	0.017	473
S2	61.8	0.144	9.26	17.8	0.017	731
S3	11.1	0.121	11.55	5.8	0.033	399
S4	6.8	0.060	8.44	9.6	0.033	360

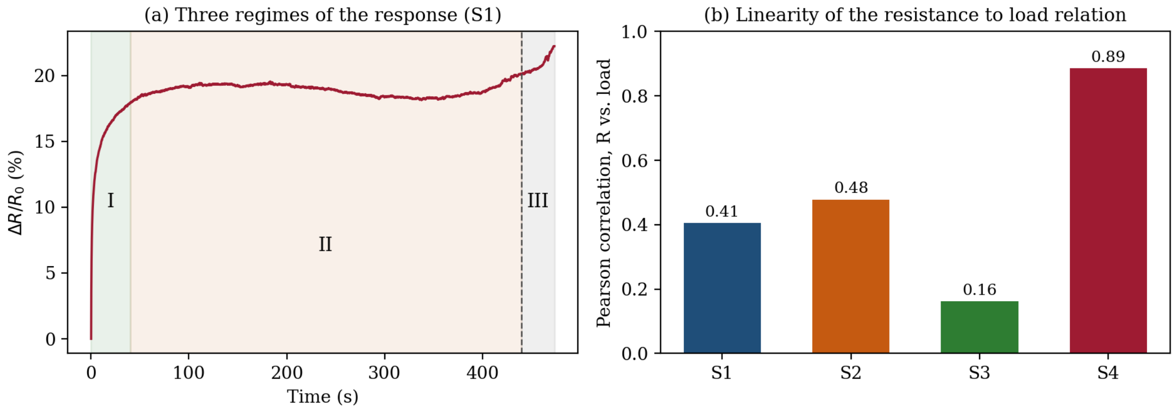

Figure 3: (a) The resistance of bar S1 divided into three stages: (I) the steep early rise, (II) the flat middle, and (III) the features around failure. (b) A straight-line fit between resistance and load varies considerably between bars.

62 MΩ, most likely because the nanotubes are not dispersed identically in each bar. This is the reason the analysis uses $\Delta R/R_0$ rather than the raw resistance.

- **The loading rate differs between bars.** Bars S1 and S2 were loaded at about 0.017 mm/s and S3 and S4 at about 0.033 mm/s, so the same event occurs at different times in different tests. This affects the choice of prediction target, discussed in Section 5.

4.1 Why a fixed formula is insufficient

Figure 3 explains why a single fixed relation will not work. The resistance passes through three clear stages (Figure 3(a)), and a straight-line fit between resistance and load is weak and varies considerably between bars, with the Pearson correlation ranging from 0.16 to 0.89 (Figure 3(b)). The relationship is non-linear, it depends on the loading history, since the same resistance can correspond to different states before and after damage, it flattens in the middle of the test, and it changes from bar to bar. A model that learns the pattern from the data, and in particular one that reads the resistance over time, is therefore the appropriate approach.

The central question is therefore whether a model can detect damage and predict the load peak from the resistance signal, and how much early warning, in seconds, it can provide.

5 Methods

This section summarises the data pipeline, the prediction target, the evaluation scheme, and the data-augmentation policy. The implementation lives under `code/src/`, and the design reasoning is recorded in the project document `PROJECT_PLAN.md`, whose relevant sections are referenced below.

5.1 Prediction target

Because the loading rate differs between bars, the time to failure is not comparable across tests. The primary target is therefore the *deflection to failure* (DTF): how much further the bar can be bent before it fails, that is, the remaining deflection in millimetres until the load peak,

$$\mathrm{DTF}(t) = \max\big(D_{\mathrm{peak}} - D(t),\, 0\big), \tag{6}$$

where $D(t)$ is the current deflection and D_{peak} is the deflection at the load peak. The deflection to failure does not depend on how fast the bar is loaded, so it is comparable across the four tests; at inference time it can be divided by the known cross-head speed to give a warning in seconds (project document, decision D2). Because the current deflection is measured live, predicting the deflection to failure amounts to inferring the failure deflection D_{peak} of an unseen bar from the shape of its resistance signal. The model also produces auxiliary estimates of load, deflection, and a four-class damage stage (early, elastic-rising, pre-failure, post-failure), which serve both as a regulariser and as virtual sensing (decision D4).

5.2 Data pipeline and causal features

The pipeline (`pipeline.py`) parses the UTM and source-meter files, resamples both onto a common 12 Hz grid, computes $\Delta R/R_0$ with R_0 taken as the median resistance over the first second, and derives the DTF, time to failure, load, deflection and stage labels (`labels.py`). Every feature is causal: at time t it uses only data at or before t (project document, §4.5), so that the system is deployable in real time. This is enforced by a unit test (`test_causality.py`) that randomises the signal after a time t and confirms that the features at or before t are unchanged.

The feature set (`engineering.py`) contains 14 features: the instantaneous $\Delta R/R_0$; its first and second derivatives with respect to deflection, computed with a one-sided causal Savitzky-Golay operator normalised by the known loading rate; rolling mean and standard deviation over deflection windows of 1, 3 and 5 mm; the cumulative $\Delta R/R_0$; a causal measure of the deflection since the first significant resistance jump; a count of step events; and the static loading rate and $\log R_0$.

5.3 Evaluation: leave-one-specimen-out

The quantity we care about is how well the model does on a bar it has never seen. Every model is therefore evaluated by *leave-one-specimen-out* (LOSO) cross-validation: the model is trained on three bars and tested on the fourth bar it has never seen, and the roles are rotated so that each of the four bars is the test exactly once. Averaging the four results is the only honest estimate of how the model would perform on a genuinely new bar (decision D5, §4.1.1), implemented in `loso.py`. Within each fold, the settings and the early-stopping point are chosen on a further

Table 2: LOSO deflection-to-failure error (mm) for each held-out bar. Lower is better.

Model	S1	S2	S3	S4	Mean
Linear regression (floor)	0.88	0.85	2.46	2.87	1.77
LightGBM (tuned)	1.27	1.05	2.10	1.60	**1.51**

held-out training bar rather than on a random split inside a bar, because neighbouring samples in a time series are almost identical and a random split would let the model in effect see the answer. A random within-bar split is never used for the test set.

5.4 Data-augmentation policy

For the sequence models the effective dataset is enlarged with five augmentation operators (`augmentations.py`), each chosen to preserve the physics (§4.2): random window crops, which is how a sequence model consumes the series; mild time-warping of ±10%, which teaches the model that the shape of the response matters more than its absolute timing; additive Gaussian noise on $\Delta R/R_0$ with a standard deviation of 0.1 to 0.3%, matched to the sensor noise; a baseline-R perturbation, which tells the model that a shift in the assumed R_0 is not informative while the curve shape is; and a cautious same-stage mixup used only late in training. Operators that would corrupt the signal, such as time reversal or random gain, are deliberately excluded.

6 Results

The models were built in three stages: classical baselines, sequence models, and a calibrated alarm. All quantities below are reported under LOSO with a fixed random seed of 42.

6.1 Classical baselines

The baseline is a gradient-boosted tree ensemble (LightGBM), tuned by an inner LOSO loop inside each fold, with linear regression as a floor; both are implemented in `baselines.py`. Table 2 reports the deflection-to-failure accuracy. LightGBM reaches a mean deflection-to-failure error of 1.51 mm, which is small relative to the 7 to 12 mm of deflection these bars sustain. Figure 4 shows the predicted and true deflection to failure on each held-out bar; the predictions track the falling line to failure, with the largest error on S3, the bar with the weakest resistance response.

6.2 Sequence models

Small causal sequence models were then built (`sequence.py`): a temporal convolutional network (TCN) and a long short-term memory (LSTM) network, each with about 20 k parameters, the multi-task heads, and the augmentation policy above. Both are neural networks that read the resistance history and are causal by construction. To make best use of only three training bars per fold, each fold uses an ensemble of three such models whose predictions are averaged (`train_sequence.py`). Table 3 reports the ablation.

The result is that no sequence model reaches the target of a 15% improvement over the baseline. The LSTM with augmentation is the best sequence model, at 1.49 mm, and marginally

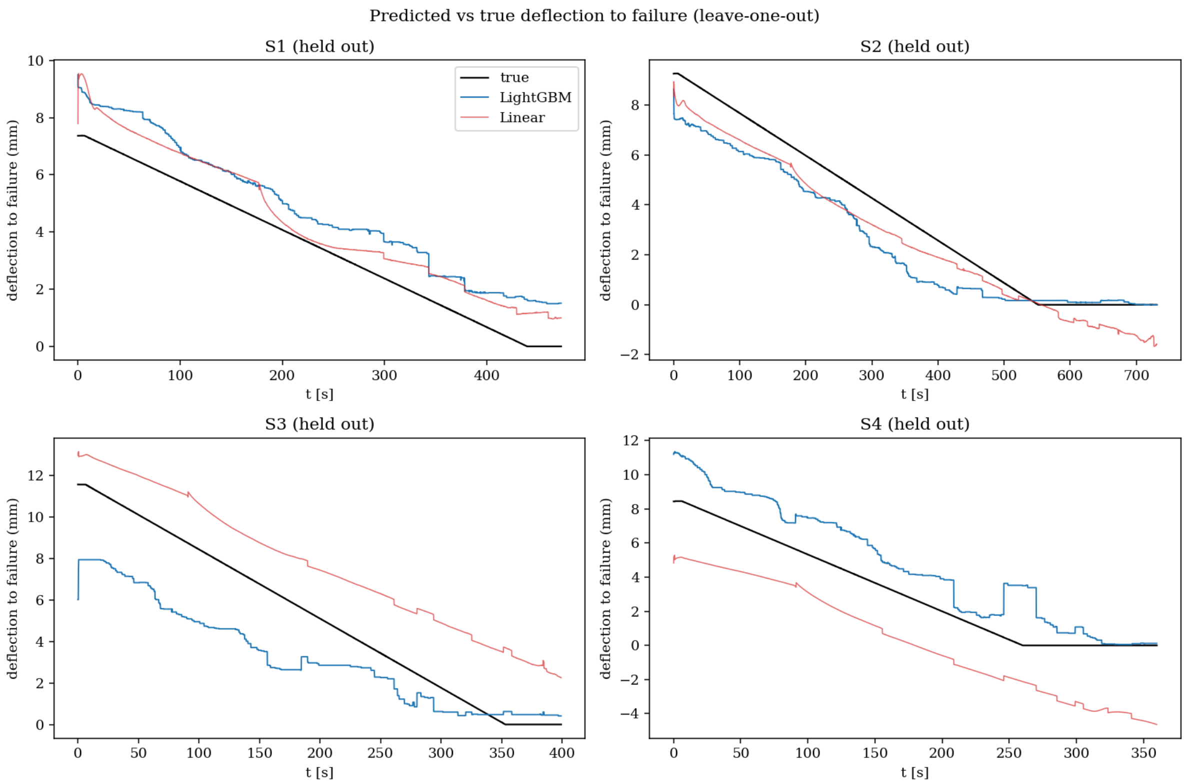

Figure 4: LightGBM (blue) and linear (red) deflection-to-failure predictions against the true value (black) for each held-out bar, under LOSO.

exceeds LightGBM, while the temporal convolutional network is worse and augmentation does not help it. With only four bars and an eightfold spread in baseline resistance, the gradient-boosted trees remain strongly competitive, which is a known outcome on small, structured datasets (decision D6). The LSTM is retained only as confirmation; the calibrated alarm below is built on the simpler and equally accurate LightGBM model.

6.3 Calibration and the early-warning alarm

The final stage converts the deflection-to-failure prediction into a calibrated alarm (`calibration.py` and `alarm.py`). Instead of a single number, the model is trained to predict a range: the 10th, 50th and 90th percentile of the deflection to failure (this is quantile regression). The alarm then uses the conservative upper bound rather than the point estimate, so that it errs on the side of caution.

A clear effect appears at this stage. Because DTF is nearly deterministic within a single bar, the raw quantile heads collapse onto one another, and the 90th-percentile band covers only about 61% of the true values on held-out bars instead of the intended 90%. The genuine source of uncertainty is the bar-to-bar variation in D_{peak}, which pooled quantile regression cannot express. This is corrected with a cross-conformal adjustment that inflates the upper bound using the exceedance measured on an inner held-out bar, restoring the coverage to 0.88 (Figure 6).

The alarm fires when the conformal upper bound on DTF stays below a threshold τ for N consecutive samples. Sweeping over (τ, N) gives the operating curve in Figure 7. The default policy selects the point that maximises the median warning time subject to recall at least 0.75

Table 3: Sequence-model ablation: mean deflection-to-failure error (mm) under LOSO, and the change relative to the LightGBM baseline of 1.51 mm.

Configuration	Mean error (mm)	vs. baseline
TCN, with augmentation, W=2 mm	1.86	−23%
TCN, no augmentation, W=2 mm	1.77	−17%
TCN, with augmentation, W=1 mm	1.81	−20%
TCN, with augmentation, W=5 mm	1.87	−24%
LSTM, with augmentation, W=2 mm	**1.49**	+1%
LightGBM baseline	1.51	

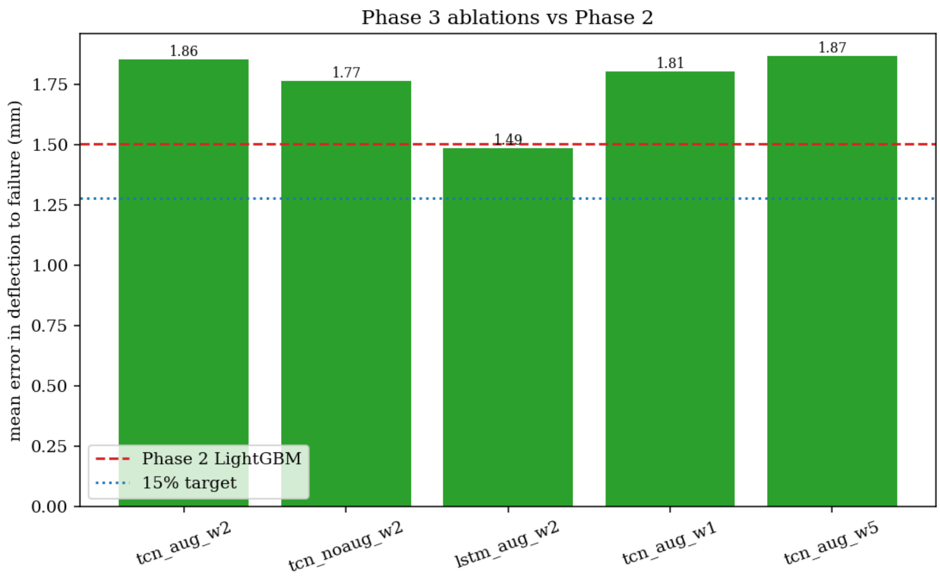

Figure 5: Sequence-model ablation against the baseline (dashed) and the 15% improvement target (dotted). Only the LSTM with augmentation reaches the baseline.

and precision at least 0.8. This gives $\tau = 5$ mm and $N = 3$ samples, which on bars the model has never seen achieves precision 1.00 and recall 1.00, with every bar caught before failure and no false alarms, and a median warning of 73.9 seconds, with per-bar warnings of 56, 279, 92 and 51 s. Figure 8 shows the calibrated band and the alarm time on each held-out bar. This meets the project's success criteria for the alarm (§7 of the project document).

7 Limitations

The scope of the conclusions is deliberately limited.

- **Only four bars.** Four bars give only four leave-one-out test points, so the uncertainty on every reported quantity is wide. The perfect precision and recall of the alarm should be read as success on all four available bars, not as a population-level guarantee.
- **Rate variability.** The bars were loaded at two different speeds. The deflection-to-failure target and the rate input mitigate this, but it remains a genuine confound.
- **Baseline-resistance spread.** The eightfold spread in R_0 is the main reason the sequence

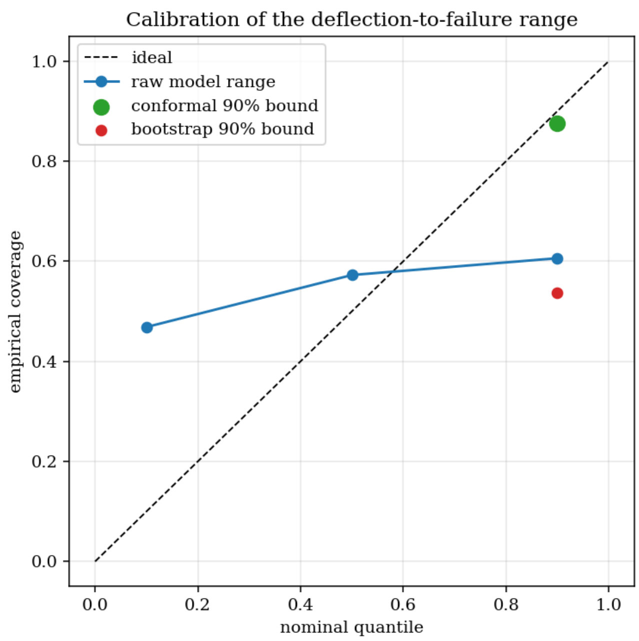

Figure 6: Calibration of the deflection-to-failure intervals under LOSO. The raw quantile heads lie far from the diagonal and are badly under-covered, while the conformal upper bound restores the intended 90% coverage.

models do not beat the trees: a held-out bar with a much weaker signal is genuinely out of distribution.

- **Two-wire measurement.** The recorded resistance includes lead and contact resistance. This is realistic for a deployed sensor but is not the cleanest signal; a four-wire measurement would be cleaner.
- **Single material and geometry.** The results do not extend to other composites, geometries, or loading modes.

8 Reproducibility

Every quantity in this report is reproducible from the raw UTM and source-meter files. The code is under `code/`, with `make` targets that run from that directory:

- `make pipeline` builds the tidy 12 Hz dataset;
- `make features` computes the causal features and labels;
- `make test` runs the causality unit test;
- `make phase2` runs the classical LOSO baselines and scorecard;
- `make phase3` runs the sequence models, augmentation and ablation;
- `make phase4` runs the calibration and the alarm operating curve;
- `make report` compiles this report, and `make all` runs the whole pipeline end to end.

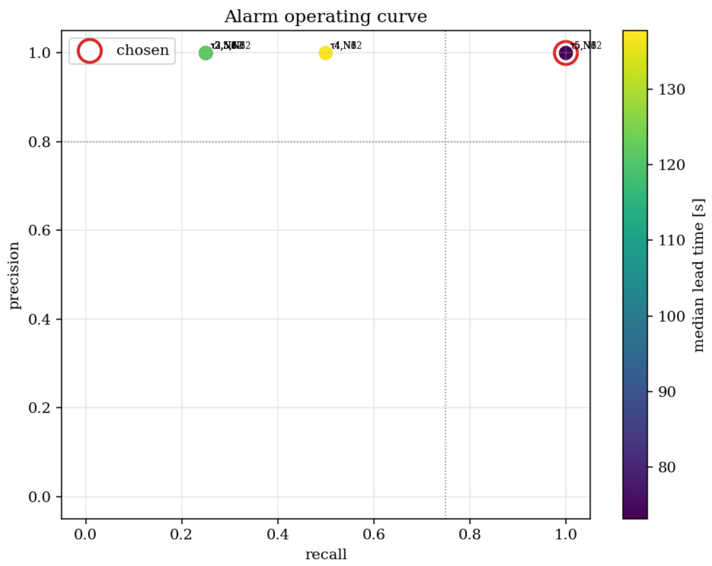

Figure 7: Alarm operating curve over (τ, N), coloured by median warning time. The selected point (circled) has precision 1.0 and recall 1.0.

All random seeds are fixed at 42 and recorded in the scorecards. Intermediate data is stored as Parquet under `data/processed/`, and the figures and per-stage scorecards under `reports/`. The gradient-boosted-tree models require the system `libomp` library.

9 Conclusion

Four CNT-filled glass-fibre bars were bent to failure while their electrical resistance was recorded. The resistance rises sharply with the first strains, flattens through the middle of the test, and shows distinct features around failure, tracking the damage in the matrix because the glass fibres carry no current. The relationship between resistance and mechanical state is non-linear, history-dependent and specific to each bar, so a learned model that reads the signal over time is the appropriate tool.

Working entirely under leave-one-specimen-out evaluation, a gradient-boosted model predicts the deflection to failure of an unseen bar to about 1.5 mm. Small sequence models, even with physics-aware augmentation, do not improve on this baseline by a meaningful margin, which reflects honestly the small number of bars available. Building on the baseline, a conformally calibrated alarm provides a defensible early-warning system: on every held-out bar it fires before failure, with a median warning of about 74 seconds and no false alarms. The principal constraint on a stronger claim is the number of specimens, and the natural next step is to test more bars, ideally with a four-wire resistance measurement.

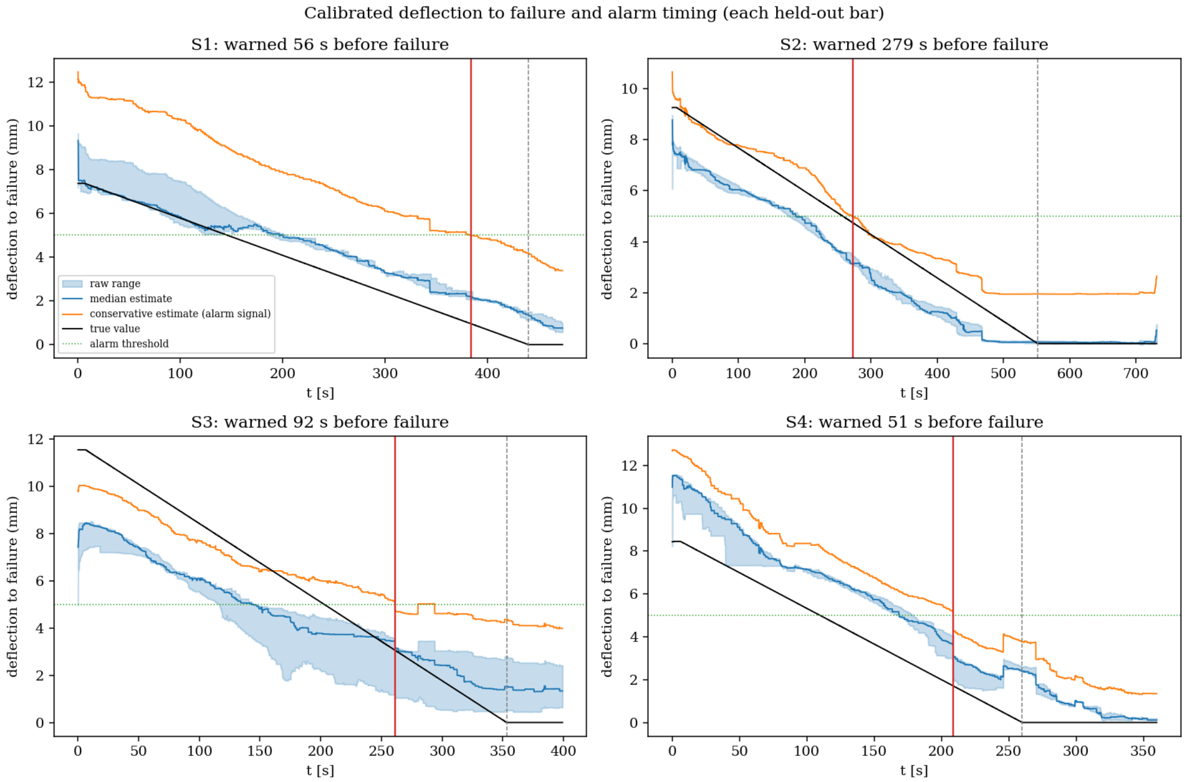

Figure 8: Calibrated deflection to failure (the conservative 90th-percentile upper bound, in orange) and the alarm time (red) against failure (grey) for each held-out bar. On all four bars the alarm fires before failure.